

Lisa Alazraki

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I am a PhD student in machine learning and natural language processing at Imperial College London. My research aims to diagnose and improve the reasoning capabilities of large language models, with a focus on robustness, compositionality, and transferability.

EDUCATION

Imperial College London , <i>PhD Computing</i>	Supervisor: Marek Rei	2023-2027
Imperial College London , <i>MSc Computing (AI and Machine Learning)</i>	Classification: Distinction	2020-2021
The Open University , <i>Grad. Cert. Theoretical Statistics and Probability</i>	Classification: Distinction	2019-2020
The Open University , <i>BSc (Hons) Computing & IT and Mathematics</i>	Classification: 1st Class	2015-2019

EXPERIENCE

Allen Institute for AI (Ai2) , <i>Research Intern</i> • Seattle, USA	Jun - Sep 2026
Manager: Varsha Kishore. Team: Asta – AI for Science.	
Meta , <i>Research Scientist Intern</i> • London, UK	Jun - Dec 2025
- Manager: Akhil Mathur. Team: Post-training Reasoning, Agents. - Studied how LLM agent capabilities scale as a function of FLOPs and completion time horizon. - Designed a marketplace-inspired adaptive multi-agent system that coordinates small agents to push the accuracy/cost Pareto frontier. - First-authored a paper accepted at ICML 2026.	
Cohere , <i>Research Intern</i> • London, UK	Jun - Dec 2024
- Manager: Max Bartolo. Team: Command Post-training. - Developed a reinforcement learning pipeline for reverse engineering human preferences that boosts LLM-as-a-judge evaluation. - Showed that LLMs attain higher accuracy without explicit corrective feedback, through implicit in-context learning. - First-authored two papers published at NeurIPS 2025 and EMNLP 2025 respectively.	
Google , <i>Research Intern</i> • Amsterdam, Netherlands	Jun - Sep 2023
- Manager: Thomas Mensink. Team: Perception. - Developed a model-ensembling framework for knowledge-intensive VQA, outperforming SOTA by 5% on Encyclopedic-VQA. - First-authored a paper presented at ICBINB at NeurIPS 2023, published in PMLR.	
Google , <i>Student Researcher</i> • London, UK	Oct - Dec 2022
<i>Research Intern</i> • Zurich, Switzerland	Jun - Sep 2022
- Manager: Hamza Harkous. Team: Applied Privacy Research. - Built a retrieval-augmented generation pipeline for user issue classification deployed to production, with 10× recall improvement. - Granted a global patent as co-inventor of the overall system.	

SELECTED PAPERS

Repetition Mismatch: Why Data Mixture Experiments Don't Scale and How to Fix Them , <i>EMNLP 2026</i>	2026
K. Zhou, L. Alazraki, K. Cao, M. Rei	
Scaling Small Agents Through Strategy Auctions , <i>ICML 2026</i>	2026
L. Alazraki, W. F. Shen, Y. Bachrach, A. Mathur	
Rethinking Rubric Generation to Improve LLM Judge and Reward Modeling for Open-ended Tasks , <i>in review</i>	2026
W. F. Shen, X. Qiu, C. Whitehouse, L. Alazraki, S. Goel, F. Barbieri, T. Willi, A. Mathur, I. Leontiadis	
AgentCoMa: A Compositional Benchmark Mixing Commonsense and Mathematical Reasoning , <i>ACL 2026 (Oral)</i>	2025
L. Alazraki, L. Chen, A. Brassard, J. Stacey, H. A. Rahmani, M. Rei	
Improving the OOD Performance of Closed-Source LLMs Through Strategic Data Selection , <i>EACL 2026 Findings</i>	2025
J. Stacey, L. Alazraki, A. Ubhi, B. Ermis, A. Mueller, M. Rei	
Reverse Engineering Human Preferences with Reinforcement Learning , <i>NeurIPS 2025 (Spotlight)</i>	2025
L. Alazraki, T. Yi-Chern, J. A. Campos, M. Mozes, M. Rei, M. Bartolo	
No Need for Explanations: LLMs Can Implicitly Learn from Mistakes In-context , <i>EMNLP 2025 (Oral)</i>	2025
L. Alazraki, M. Mozes, J. A. Campos, T. Yi-Chern, M. Rei, M. Bartolo	
Enhancing LLM Robustness to Perturbed Instructions: An Empirical Study , <i>ICLR 2025 BuildingTrust</i>	2025
A. Agrawal*, L. Alazraki*, S. Honarvar, M. Rei (*Equal contribution)	
Meta-reasoning Improves Tool Use in Large Language Models , <i>NAACL 2025 Findings</i>	2024
L. Alazraki, M. Rei	
How (not) to ensemble LVLMs for VQA , <i>NeurIPS 2023 ICBINB / PMLR 239</i>	2023
L. Alazraki, L. Castrejon, M. Dehghani, F. Huot, J. Uijlings, T. Mensink	

AWARDS

Placement Award , <i>Alan Turing Institute</i>	2024
Selected as one of ~30 resident PhD students for the Institute's Enrichment Scheme 2024/25.	
ICC Poster Competition First Prize , <i>Imperial College London</i>	2024
Awarded for best poster presentation at the Imperial Computing Conference.	
IET Postgraduate Prize , <i>Institution for Engineering and Technology</i>	2024
Awarded for 'excellence in postgraduate research' at the IET Postgraduate Research Awards.	
Best Student Paper Award , <i>Institute of Electrical and Electronics Engineers</i>	2023
Awarded at the IEEE International Conference on Cognitive Machine Intelligence.	
Distinguished Dissertation Award , <i>Imperial College London</i>	2021
Awarded for 'outstanding MSc dissertation in terms of technical achievement and presentation'.	
Faculty Commendation , <i>The Open University</i>	2017
Awarded for achieving one of the highest results out of ~500 students in Algorithms, Data Structures and Computability.	
Leslie Walshaw Award , <i>The Open University</i>	2016
Awarded for achieving the highest overall grade of all students in the Greater London region in mathematics module MST125.	

LEADERSHIP AND COMMUNITY

Supervision	Conceived 10+ MSc/MEng thesis specifications and co-supervised 15 students across machine learning and NLP
Mentorship	Hold regular office hours advising applicants on graduate admissions, particularly from non-traditional backgrounds
Organisation	Co-organise Imperial's NLP reading group hosting internal and external speakers; support committee at the ICC Conference
Teaching	Graduate teaching assistant across five courses spanning computer science, machine learning, deep learning, and NLP
Reviewing	Reviewer for ACL Rolling Review, COLING, COLM, ICML (Gold Reviewer), NeurIPS, TMLR, as well as various workshops

SELECTED TALKS

Invited

Reverse Engineering Human Preferences with Reinforcement Learning, <i>Lancaster AI, Lancaster University</i>	2026
Reverse Engineering Human Preferences with Reinforcement Learning, <i>Breaking Topics in AI Conference, I-X Imperial</i>	2025
Benchmarking LLMs on Mixed-type Compositional Reasoning, <i>Open Science Community Lightning Talks, Cohere Labs</i>	2025

Contributed

Mixing Commonsense and Mathematical Reasoning in Real-World Scenarios, <i>NLP Special Interest Group, Alan Turing Institute</i>	2026
Pattern Matching vs. Procedural Reasoning in Language Models, <i>Imperial NLP, Imperial College London</i>	2026
Plan Auctions for Task-Aware Selection of LLM Agents Across Scales, <i>Meta Superintelligence Labs Tech Talks, Meta</i>	2025
No Need for Explanations: LLMs Can Implicitly Learn from Mistakes In-context, <i>Conference on Empirical Methods in NLP</i>	2025
Overcoming the Limitations of Autoregressive Generation, <i>Imperial NLP, Imperial College London</i>	2024
A Retrieval-Augmented Generation Pipeline for User Issue Classification, <i>T5 and Beyond Monthly Talks, Google</i>	2022

GRANTS AND SCHOLARSHIPS

IX-WAI Early Career Development Grant , <i>I-X Imperial</i>	2025
OpenAI API Researcher Access Grant , <i>OpenAI</i>	2025
Annual Grant , <i>Sir Richard Stapley Trust</i>	2024
General Fund Grant , <i>Imperial College Trust</i>	2023
IET International Travel Award , <i>Institution for Engineering and Technology</i>	2023
Annual Grant , <i>Sir Richard Stapley Trust</i>	2023
EPSRC Doctoral Scholarship , <i>Imperial College London</i>	2022
DeepMind MSc Scholarship , <i>Imperial College London</i>	2020

PATENTS

Deep Learning System for Navigating Feedback , <i>US Patent US20250272485A1, CN Patent CN119053957A, WO Patent WO2023224672A1</i>
Inventors: H. Harkous, S. T. Peddinti, R. Khandelwal, A. Srivastava, N. Taft, <u>L. Alazraki</u> . Assignee: Google LLC

TRAINING AND SKILLS

Summer schools attended	Oxford Machine Learning Summer School MLx Representation Learning & GenAI 2025 + MLx Health & Bio 2025
Programming languages	Python, TypeScript, JavaScript, Java, Lua, MATLAB/Octave, Maxima, C++, Solidity, Prolog, Unix/Bash, HTML, CSS
Libraries / frameworks	PyTorch, TensorFlow, Transformers, vLLM, TRL, NumPy, Pandas, Scikit-learn, Jinja2, Matplotlib, Plotly, WanDB